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MACHINE LEARNING

Lecture 1.10

Mistake Model of Learning (by Nick Littlestone)

We say a learner has mistake bound t, If for every sequence of challenges, learner makes atmost t mistakes.

For eg:→
their is a learner, a teacher and some datapoint (such as email). The learner has to guess whether the email is SPAM / NOT-SPAM

← DATA POINT 1

Learner "SPAM" → GUESS

← CORRECT — TEACHER

← EMAIL 2

"NOT SPAM" →

← MISTAKE — TEACHER

For every mistake, the learner will update the mistake count by 1. And the learner will update its internal state.

So, for example →

$C = \{$MONOTONIC DISJUNCTIONS ON N VARIABLES$\}$

DOMAIN $= \{0,1\}^n$

Here ~~disjunction~~ ~~conjunction~~ means: →

① Disjunction → OR of variables.

ex: → $x_1 \vee x_2 \vee x_3$

This expression is true if atleast one variable is true

② Monotone → no negation is allowed

We can use the variables themselves, not their negations

eg: → $x_1 \vee \neg x_2$ (not allowed)

③ n variables

This means we have n boolean variables.

So $f(x) = x_1 \vee x_7 \vee x_9$

This f^n is going to be 1, if it has 1 at its 1st position or 7th position or 9th position

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Lecture 1.1.1

Q) Can you come up with a learner/algo with mistake bound at most n?

Learner will start with the monotone dist $x_1 \vee x_2 \vee \cdots \vee x_n$

Learner ← Receives 011000 ← Teacher

Learner → 1 → Teacher

← MISTAKE ←

So, this implies that
$\Rightarrow x_2$ cannot be in this unknown disj^n
$\Rightarrow x_3$ cannot be in this unknown disj^n
$\Rightarrow x_6$ cannot be in this unknown disj^n

Hence, every time we make a mistake AT LEAST one literal is eliminated. Their are at most N literals
$\Rightarrow$ # MISTAKES $\leq n$

So, even though we can be presented with 2^n challenges, this algorithm will only make one mistake.

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Lecture 1.1.2

Now instead of monotonous disjunctions lets see only about disjunctions.

$C = \alpha$ disjunctions γ

Question → How can we use the algo for monotone disjunctions to learn disjunctions?

Solution → we can do the feature expansion

$$x_1 x_2 \ldots x_n \longrightarrow \underbrace{x_1 x_2 \ldots x_n y_1 y_2 \ldots y_n}_{2n}$$

each $y_i = \overline{x_i}$

Hence, the function

0 1 1 0 $(n=4)$ ⟶ 0 1 1 0 1 0 0 1.

$$f(x_1 x_2 \ldots x_n) = x_2 \vee \overline{x_4} \vee x_7$$

(equals)

$$f(x_1 x_2 \ldots x_n y_1 y_2 \ldots y_n) = \underbrace{x_2 \vee y_4 \vee x_7}$$

Now apply monotone disjunction

So for this algo,
we have a new algo with mistake bound $\leq 2n$